

Ehsan Sharafian

 ehsan.sharafian@maine.edu

 (207) 631-6979

 LinkedIn

 GitHub

 ehsansharafian.com

 Google Scholar

PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

EDUCATION

Ph.D. in Mechanical Engineering | University of Maine, Orono, ME

Expected Dec 2027

GPA: 4.00/4.00 | Dissertation: Smart Real-Time Gait Training System

M.S. in Mechanical Engineering | Amirkabir University of Technology (Tehran Polytechnic)

Feb 2020

GPA: 3.65/4.00 | Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces

B.S. in Mechanical Engineering | University of Tehran

Jul 2017

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS
Mobile & Software	MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules
Biomechanics & Signal Processing	Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim
Robotics	Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant | Biorobotics and Biomechanics Lab, University of Maine

Sep 2023 – Present

Orono, ME

- Designed and validated a real-time, smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify level walking, stair ascent/descent, and ramp ascent/descent with 96.08% accuracy, less than 1.1 cm cumulative height error, and 112 ms latency.
- Engineered a lightweight human activity recognition system using two IMUs and machine learning techniques to detect normal locomotion with 99.1% accuracy and classify atypical stair-negotiation strategies in older adults with 93.1% accuracy and 53 ms latency.
- Developed a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry to provide real-time thigh-extension feedback for adults aged 65+, increasing stride length and walking speed by up to 30% in a single-session study with minimal cognitive demand.
- Validated wearable systems with human participants, including 40 adults aged 65+ and 2 stroke patients in a haptic feedback gait study, 20 young adults in a foot-clearance study, and 10 young adults plus 15 older adults in activity recognition experiments.
- Implemented real-time algorithms and data analysis pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Tested and demonstrated the system with 6 experienced surgeons and supported medical-student workshops through operator training, troubleshooting, and usability feedback collection.

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored **6 undergraduate students** across **5 research teams**, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the **9th International Conference on Robotics and Mechatronics**, including lab demonstrations, equipment readiness, and presentation logistics.

SELECTED HONORS

Flagship Fellowship Award, University of Maine	Sep 2024 – Sep 2025
Graduate Student Government Award for conference proposal	Mar 2025

LEADERSHIP & SERVICE

Graduate Senator , Department of Mechanical Engineering, University of Maine	Sep 2025 – Present
President , Iranian Graduate Student Association, University of Maine	Sep 2024 – Apr 2026

PUBLICATIONS

Six peer-reviewed journal publications in wearable sensing, biomechanics, rehabilitation engineering, and robotics.

1. **Sharafian, E.**, & Hejrati, B. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." *Sensors*, 2026.
2. **Sharafian, E.**, Ellis, C., & Hejrati, B. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." *IEEE Access*, 2025.
3. **Sharafian, E.**, et al. "The Effects of Real-Time Haptic Feedback on Gait and Cognitive Load in Older Adults." *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2025.
4. Noghani, M. A., **Sharafian, E.**, Sidaway, B., & Hejrati, B. "Increasing Thigh Extension with Haptic Feedback Affects Leg Coordination in Young and Older Adult Walkers." *Journal of Biomechanics*, 2025.
5. **Sharafian, E.**, Ghassabzadeh, M., & Taghvaeipour, A. "Trajectory Optimization of a Spot-Welding Robot in the Joint and Cartesian Spaces." *International Journal of Robotics and Automation*, 2023.
6. **Sharafian, E.**, Taghvaeipour, A., & Ghassabzadeh, M. "Revisiting Screw Theory-Based Approaches in the Constraint Wrench Analysis of Robotic Systems." *Robotica*, 2022.